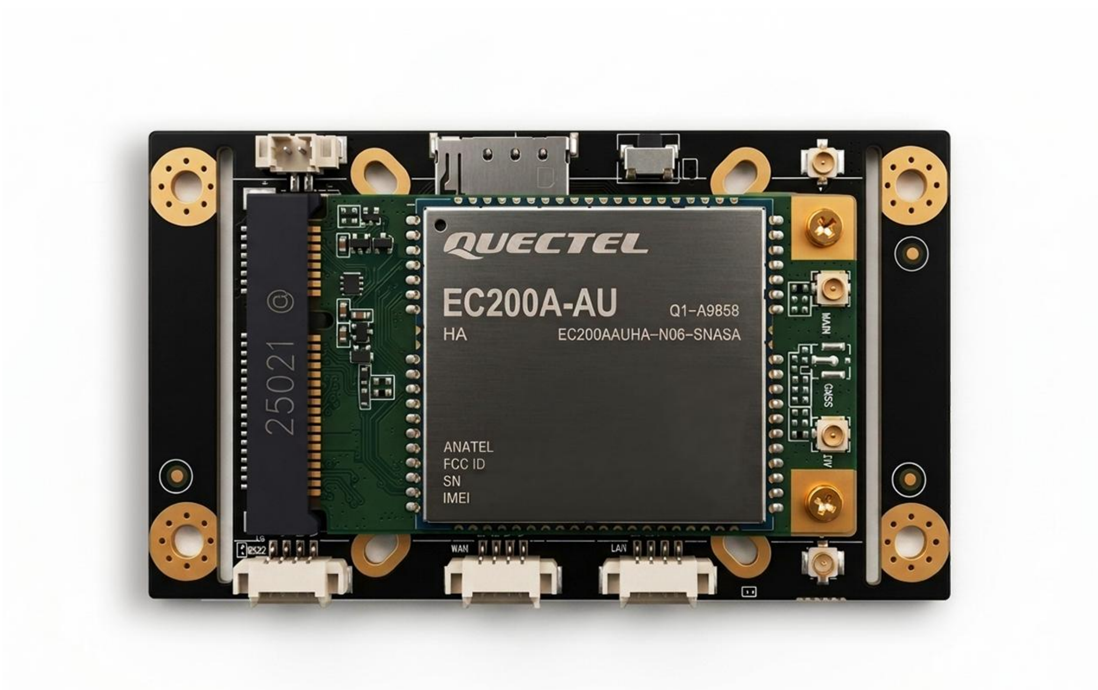
D218R 4G Router Module

Public Datasheet | Rev. 1.0

D218R is a compact LTE router module for OEM equipment, industrial gateways, remote sites, and embedded connectivity projects. It combines cellular access, dual 10/100 Mbps Ethernet, 2.4 GHz Wi-Fi, dual serial interfaces, configurable IO, and remote management support in a board-level PCBA form factor.



Product positioning	Summary
Product type	Industrial LTE router module with expansion interfaces
Target users	OEMs, system integrators, industrial IoT teams, and mobile equipment builders
Typical projects	Remote gateway, cellular failover, serial-to-IP, IO-to-MQTT, video backhaul, and private-network access
Availability	Evaluation samples and project configurations available by request

1. Key Features

- Board-level industrial router module for OEM integration and field equipment connectivity.
- Cellular access path based on selected LTE module and regional project configuration.
- Two 10/100 Mbps Ethernet interfaces with WAN/LAN behavior controlled by operating mode.

- Two serial interfaces for transparent transmission, Modbus, MQTT, command mode, GPS/BeiDou data, HTTP POST, or custom protocol conversion.
- 2.4 GHz 802.11 b/g/n Wi-Fi with AP and client/relay use cases.
- Three configurable 3.3 V IO lines and 50Pin expansion connector for extended interface access.
- Remote management platform support for status, Web UI entry, terminal access, OTA workflow, and project integration.

2. Typical Applications

Application	How D218R is used
Industrial router module	Embedded inside customer equipment as a compact LTE router, Ethernet gateway, Wi-Fi access point, and remote access node
Serial and IO gateway	Connects PLCs, sensors, meters, controllers, and field devices to MQTT, HTTP, TCP/UDP, Modbus, or customer platforms
Mobile and field equipment	Provides cellular access, Wi-Fi connectivity, VPN/private networking, and remote diagnostics for distributed field assets
OEM remote operations	Supports remote login, Web UI access, terminal sessions, status reporting, OTA upgrade, and platform-based device maintenance

3. Core Specifications

Item	Public datasheet value
Cellular module	Reference configuration may use Quectel EC20 / EC200 series LTE module; regional LTE module options may vary by project
Cellular standard	3G / 4G LTE; final cellular standard support depends on selected module, firmware, carrier requirements, and target region
4G bands	Reference LTE configuration: FDD LTE B1/B3/B8; TDD LTE B38/B39/B40/B41; Final band support depends on selected module and region
Ethernet	2 x 10/100 Mbps interfaces through 4-pin 1.25 mm connectors; WAN/LAN roles depend on operating mode
Wi-Fi	2.4 GHz 802.11b/g/n 2T2R up to 300 Mbps

Serial / IO	UART1 default TTL with optional RS232 customization; UART2 default RS485 with optional TTL mode; 3 configurable 3.3 V IO channels with I2C-style use by project
SIM	Nano SIM slot; SIM hot-swap is not supported
Antenna	2 x LTE antenna interfaces by default; 2 x 2.4 GHz Wi-Fi antenna interfaces; GPS antenna interface depends on model/configuration
Power input	7-15 V, recommended supply 12 V / 1 A; 5 V input support must be confirmed by project configuration
Typical power	Less than 3 W listed under full-load reference conditions; provide at least 5 W power budget for cellular peak load
Operating temperature	Supports -40 °C cold start and long-term operation up to +70 °C
Board outline	73.4 mm × 42.0 mm × 12.9 mm with locating plate; 58.0 mm × 42.0 mm × 12.9 mm after removing locating plate. Verify against mechanical drawing before enclosure design
50-pin expansion	Dual-row 1.0 mm pitch connector for expanded Ethernet, serial, SIM, reset, indicators, IO, and power access

4. Product Views and Interface Overview

This section provides front and rear product views, interface layout, and mechanical dimension references for appearance identification, interface position confirmation, and preliminary mechanical integration evaluation. Complete pin definitions, electrical thresholds, grounding/shielding design, protection design, connector height, enclosure clearance, and thermal conditions should be further confirmed during the project engineering phase.

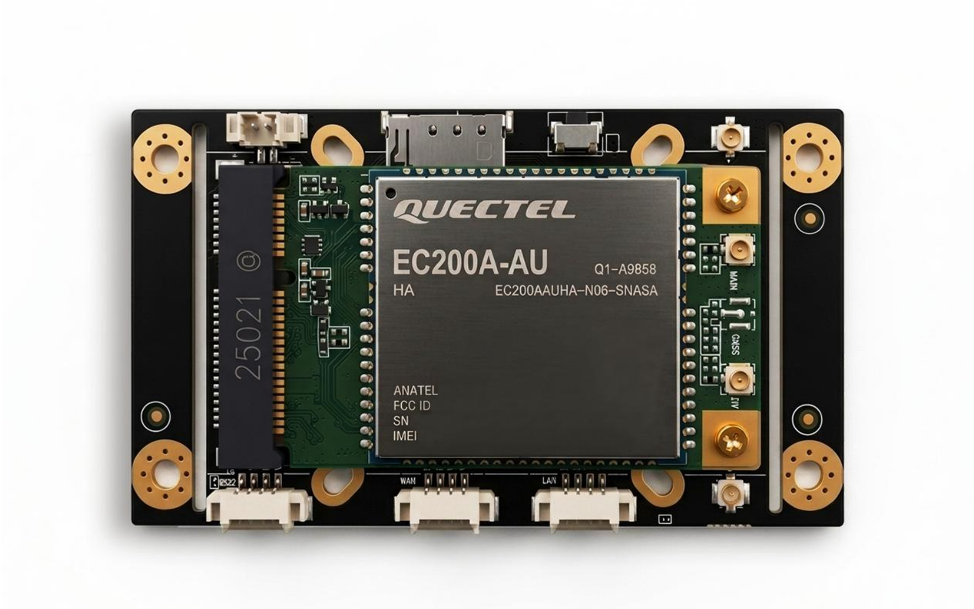


Figure 1. Front product view

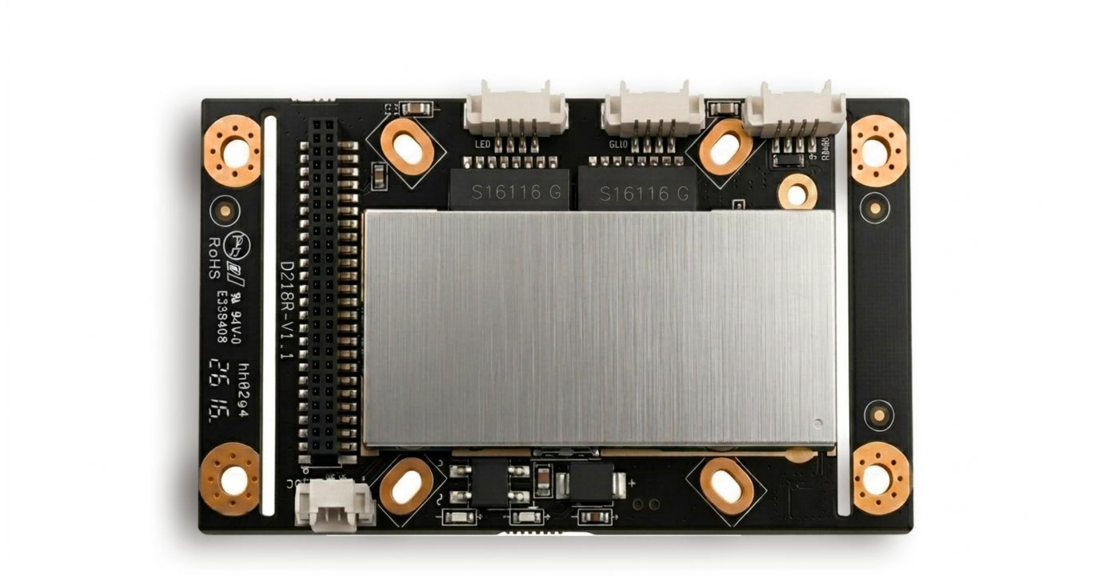


Figure 2. Rear product view

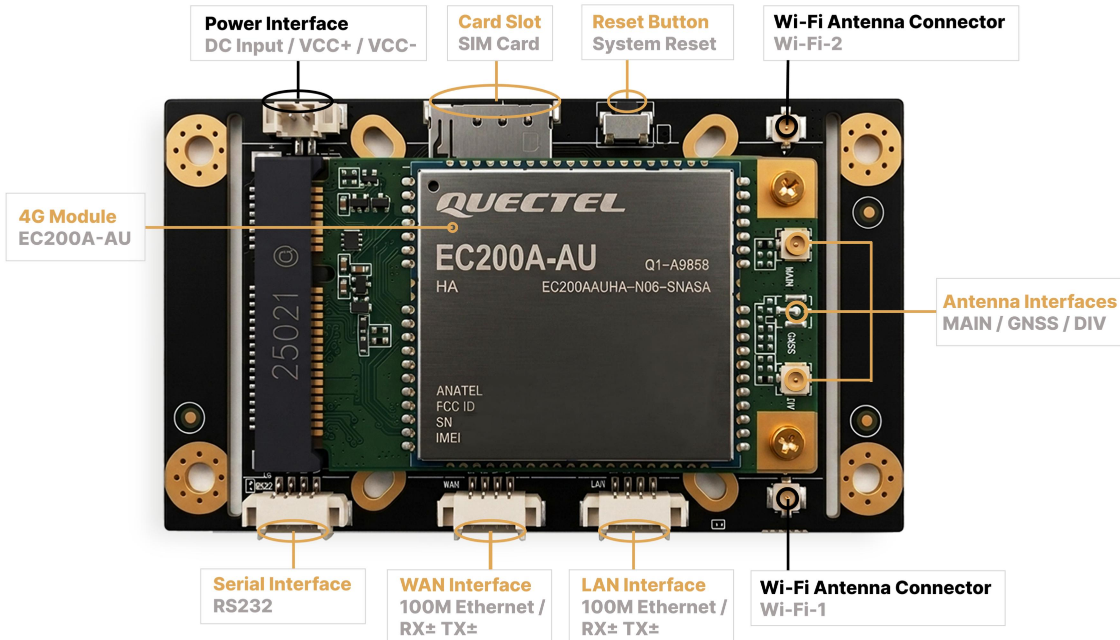


Figure 3. Front interface and major component layout

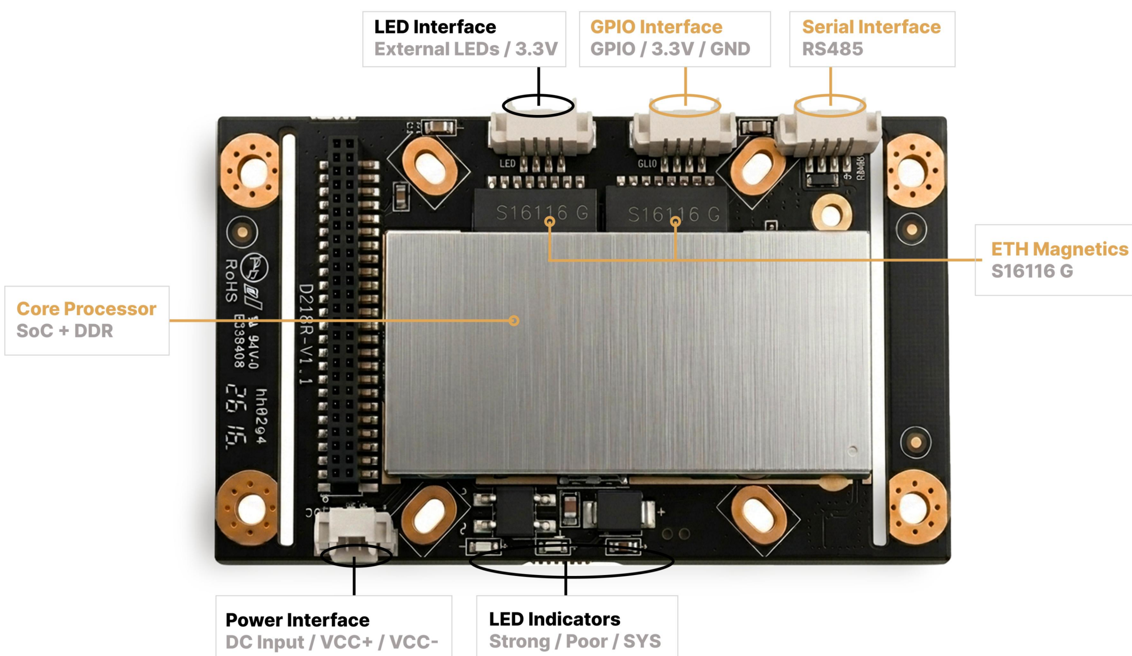


Figure 4. Rear interface and major component layout

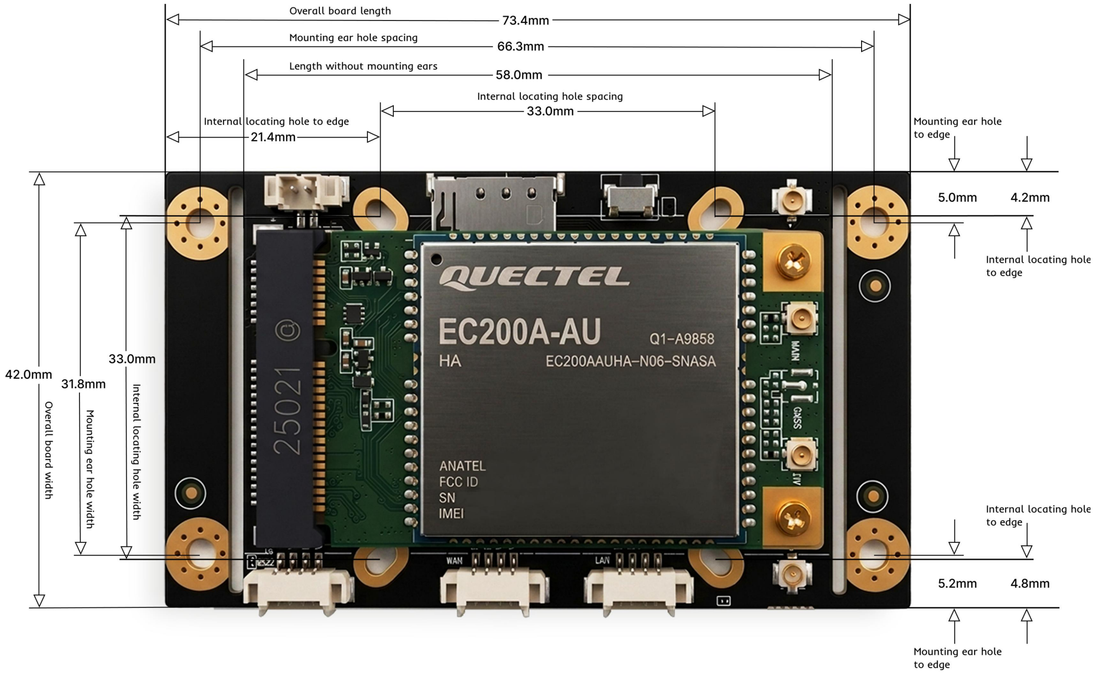


Figure 5. Front mechanical dimensions and mounting reference

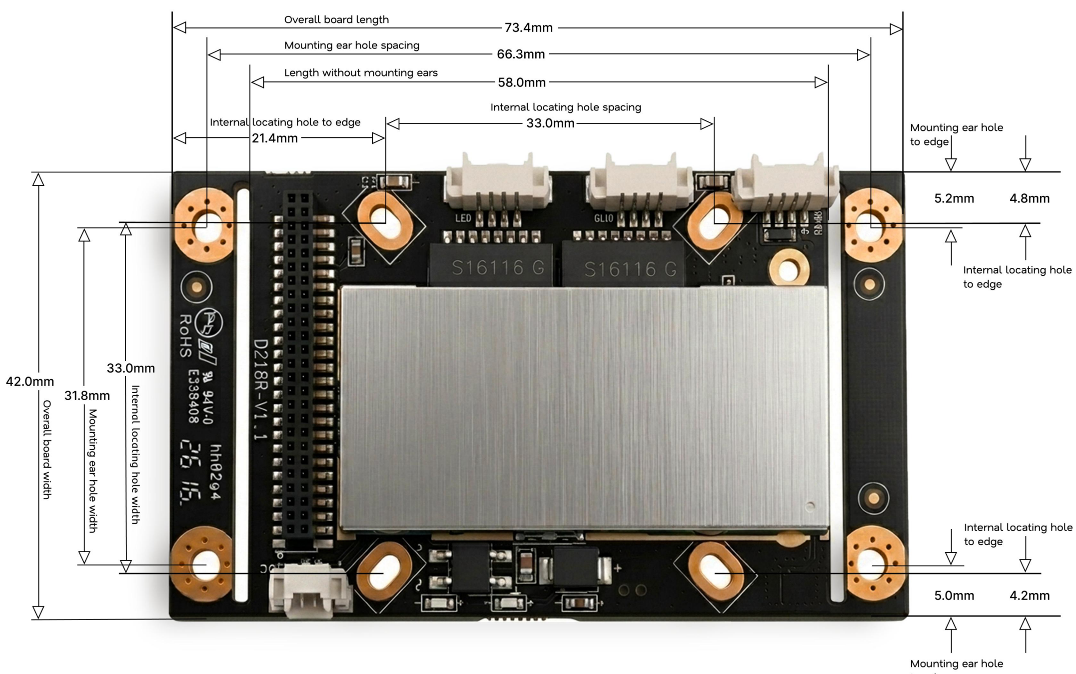


Figure 6. Rear mechanical dimensions and mounting reference

Interface group	Public summary
Power	Horizontal and vertical 2-pin 1.25 mm power input positions are available; use the defined input position for the project design
Ethernet	WAN and LAN 10/100 Mbps interfaces are available on the module; additional Ethernet signals may be accessed through the expansion connector with proper transformer design
Serial / IO	UART1, UART2, and configurable IO lines support serial data, field device connection, industrial control, and project-defined interface behavior
SIM / Reset / Indicators	Nano SIM, reset, system indicator, signal indicators, and related signals support field installation and maintenance workflows
Expansion connector	The 50-pin connector can provide extended Ethernet, serial, SIM, reset, indicator, IO, and power access; detailed pinout is provided during project engineering review

Engineering Materials Available on a Project Basis

During the project evaluation phase, detailed pin definitions, electrical thresholds, connector height, component height, DXF/STEP reference files, protection design notes, and enclosure integration evaluation support can be provided as needed.

5. Product Variants

Variant	Description
D218R Standard	LTE router module with UART1 TTL, UART2 RS485/TTL configurable, Ethernet, Wi-Fi, IO, and remote management support
D218R-232	Adds RS232 configuration on UART1 for projects that require RS232 serial integration
D218R-GPS	Adds GPS positioning capability for projects that require location data; GPS antenna/interface availability should be confirmed by final configuration
D218R Core	For projects that require customer-selected LTE module integration through the onboard MiniPCIe socket; module compatibility, antenna routing, firmware support, thermal behavior, and certification scope require project review

6. Software and Remote Management

- Talonbox SkinOS embedded system platform with component-based development, software package deployment, SDK support, and project-level customization.
- Local TCP control, local HTTP status query, LAN discovery, SMS management, remote HTTP/MQTT reporting, and platform-based device management.
- Remote Web UI access, terminal access, OTA upgrade, configuration management, log review, platform API, and maintenance workflows.
- Cellular, wired WAN, Wi-Fi client/relay, bridge/repeater, and hybrid multi-WAN operation depending on firmware and project configuration.
- Serial protocol support includes transparent transmission, Modbus RTU to Modbus TCP, MQTT, command mode, GPS/BeiDou data, HTTP POST, and custom protocol conversion.
- Firewall, NAT/DMZ, port proxy, static routing, policy routing, SNMP, DDNS, UPnP, PPTP, L2TP, IPsec, GRE, OpenVPN, WireGuard, NAT traversal, and high-availability features where supported.

7. Ordering and Project Notes

Item	Public note
Evaluation sample	Available by project request; target region, carrier, antenna plan, interface requirements, and intended application should be provided
OEM configuration	Firmware defaults, remote platform settings, serial behavior, IO definition, protocol conversion, branding, and interface options can be reviewed by project
Lead time	Eligible standard configurations may start from 7 days; final timing depends on quantity, cellular module option, certification, connector/interface configuration, and customization scope
MOQ	Sample and production MOQ should be confirmed by project
Supporting files	Mechanical drawing, interface layout, connector reference, and engineering package are available during project review

8. Certification and Configuration Notice

- Certification status, cellular module options, antenna configuration, supported frequency bands, product labels, reference images, carrier acceptance, access permissions, encryption policy, platform configuration, and delivery timing may vary by target market and project configuration.
- Module-level certification marks shown in reference images apply to the cellular module only. They do not automatically indicate final D218R board-level certification or end-product certification.
- The final customer product, including enclosure, antenna design, power input, connector/interface design, serial/IO wiring, installation method, labeling, and regional deployment requirements, may require additional engineering review or certification.

9. Contact and Project Review

To request a datasheet package, evaluation sample, OEM integration review, mechanical files, connector reference, or remote management support, please contact Talonbox with your project requirements, including target region, carrier requirements, enclosure plan, and required interfaces.

Recommended information to include

Target region, carrier requirements, expected quantity, enclosure plan, required interfaces, serial/IO requirements, and whether the project requires a standard D218R configuration or an OEM integration review.

- Email: talonbox@ashyelf.com
- Website: https://www.talonbox.com
- Inquiry form: https://www.talonbox.com/#contact

Final configuration, delivery timing, certification scope, and engineering file availability are confirmed by project.